

HandsMen Threads Salesforce Project Documentation

Prepared for Project Review

Salesforce Platform Developer Project

Data Model Automation Security Apex Testing

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This report presents the HandsMen Threads Salesforce implementation in a concise review format. Each numbered figure panel identifies the exact screenshot that should be captured from the Salesforce developer org before submission.

1 Project Overview

HandsMen Threads is a Salesforce-based solution designed for a fashion business to manage customers, products, orders, inventory, and marketing campaigns. The project centralizes business operations on a single platform while using automation to streamline customer communication, inventory management, loyalty programs, and order processing.

The implementation includes custom objects and fields, lookup and master-detail relationships, formula fields, validation rules, profiles, roles, permission sets, email templates, record-triggered and scheduled flows, Apex triggers, Batch Apex, and Scheduled Apex.

Main Business Outcomes

- Automatically send confirmation emails when an order is successfully confirmed.
- Automatically update customer loyalty status based on purchase history.
- Notify the inventory team when product stock falls below the defined threshold.
- Automatically calculate order totals and deduct inventory when confirmed orders are processed.
- Handle bulk inventory updates efficiently using scheduled Apex automation.

Technology Used

Area	Salesforce Feature
Data	Custom objects, custom fields, formula fields, relationships
Automation	Record triggered flows and scheduled flow
Development	Apex classes, triggers, handler pattern, Batch Apex, Scheduled Apex
Security	Profiles, roles, users, permission sets, object access
Communication	Classic email templates and email alerts

2 Objectives and Implementation Scope

2.1 Project Objectives

- Design a structured Salesforce data model to manage customers, products, orders, inventory, and marketing campaigns.
- Ensure accurate and consistent customer, order, and inventory data through validation rules and controlled data entry.
- Automate key business processes such as order confirmations, low-stock notifications, and customer loyalty updates.
- Implement role-based access and permissions to ensure users can access only the records relevant to their responsibilities.
- Use Apex to handle order validation, order total calculation, inventory deduction, and bulk inventory processing.

2.2 Current Implementation Scope

Area	Included Implementation	Review Evidence
Data model	Five custom objects with business fields and relationships	Object Manager and Schema Builder
Data quality	Formula fields and validation rules	Field and validation rule screens
Security	Profile, role hierarchy, users, permission set	Setup security screens
Automation	Order, stock, and loyalty flows	Flow Builder canvases
Development	Handler, triggers, batch, and scheduler	Apex class and trigger screens
Testing	Sample records, error messages, updates, and emails	Lightning records and test results

3 Requirement Analysis and Data Model

The HandsMen Threads data model is designed to clearly separate **master data from transactional data**. Customer and product records maintain reusable business information, while order records capture customer transactions. Inventory records track product stock levels, and marketing campaign records support promotional activities and customer engagement. This structured approach improves data organization, consistency, and scalability across the application.

Custom Object	Key Fields	Business Purpose
HandsMen Customer	Name, First Name, Last Name, Email, Phone, Loyalty Status, Total Purchases	Stores customer identity, contact, and loyalty details.
HandsMen Product	Name, SKU, Price, Stock Quantity	Stores the product catalogue and pricing.
HandsMen Order	Order Number, Status, Quantity, Total Amount, Customer, Product	Captures customer purchases and order status.
Inventory	Inventory Number, Warehouse, Stock Quantity, Product	Tracks stock levels for each product.
Marketing Campaign	Campaign Number, Start Date, End Date, Customer	Stores promotional campaigns and customer association.

Setup

Home

Object Manager

SETUP > OBJECT MANAGER

HandsMen Customer

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Search Layouts

List View Button Layout

Fields & Relationships

11 Items, Sorted by Field Label

Quick Find

New

Deleted Fields

Field Dependencies

Set History Tracking

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		
Email	Email__c	Email		
FirstName	FirstName__c	Text(20)		
Full_Name__c	Full_Name__c_c	Formula (Text)		
HandsMen Customer Name	Name	Text(80)		✓
Last Modified By	LastModifiedById	Lookup(User)		
LastName	LastName__c	Text(20)		
Loyalty Status	Loyalty_Status__c	Picklist		

Figure 1 HandsMen Customer object fields including email phone loyalty status and total purchases

Setup

Home

Object Manager

SETUP > OBJECT MANAGER

HandsMen Product

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Search Layouts

List View Button Layout

Fields & Relationships

8 Items, Sorted by Field Label

Quick Find

NewDeleted FieldsField DependenciesSet History Tracking

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		
HandsMen Product Name	Name	Text(80)		✓
Last Modified By	LastModifiedById	Lookup(User)		
Order	Order__c	Lookup(HandsMen Order)		✓
Owner	OwnerId	Lookup(User,Group)		✓
Price	Price__c	Currency(18, 0)		
SKU	SKU__c	Text(18)		
Stock Quantity	Stock_Quantity__c	Number(18, 0)		

Figure 2 HandsMen Product object fields including SKU price and stock quantity

Setup

Home

Object Manager

SETUP > OBJECT MANAGER

HandsMen Order

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Search Layouts

List View Button Layout

Fields & Relationships

10 Items, Sorted by Field Label

Quick Find

NewDeleted FieldsField DependenciesSet History Tracking

Customer	Customer__c	Lookup(HandsMen Customer)	✓	
Customer_Email	Customer_Email__c	Email		
HandsMen Order Name	Name	Auto Number	✓	
HandsMen Product	HandsMen_Product__c	Lookup(HandsMen Product)	✓	
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)	✓	
Quantity	Quantity__c	Number(18, 0)		
Status	Status__c	Picklist		
Total Amount	Total_Amount__c	Number(18, 0)		

Figure 3 HandsMen Order object fields and relationships including status quantity total amount customer and product

SETUP > OBJECT MANAGER
Inventory

Details
Fields & Relationships
7 Items, Sorted by Field Label

Q, Quick Find New Deleted Fields Field Dependencies Set History Tracking

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		
Inventory Name	Name	Auto Number		✓
Last Modified By	LastModifiedById	Lookup(User)		
Product	Product__c	Master-Detail(HandsMen Product)		✓
Stock Quantity	Stock_Quantity__c	Number(18, 0)		
Stock Status	Stock_Status__c	Formula (Text)		
Warehouse	Warehouse__c	Text(18)		

Figure 4 Inventory object fields and relationship including warehouse stock quantity and related product

SETUP > OBJECT MANAGER
Marketing Campaign

Details
Fields & Relationships
7 Items, Sorted by Field Label

Q, Quick Find New Deleted Fields Field Dependencies Set History Tracking

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		
End Date	End_Date__c	Date		
HandsMen Customer	HandsMen_Customer__c	Lookup(HandsMen Customer)		✓
Last Modified By	LastModifiedById	Lookup(User)		
Marketing Campaign Name	Name	Auto Number		✓
Owner	OwnerId	Lookup(User, Group)		✓
Start Date	Start_Date__c	Date		

Figure 6 Marketing Campaign object fields and relationship including campaign dates and related customer

3.1 Relationships and Important Fields

Child Object	Relationship	Related Object	Purpose
Marketing Campaign	Lookup	HandsMen Customer	Associates a campaign with a customer.
HandsMen Product	Lookup	HandsMen Order	Connects product information to an order as documented.
HandsMen Order	Lookup	HandsMen Customer	Links each order to the purchasing customer.
Inventory	Master Detail	HandsMen Product	Makes inventory dependent on its product.

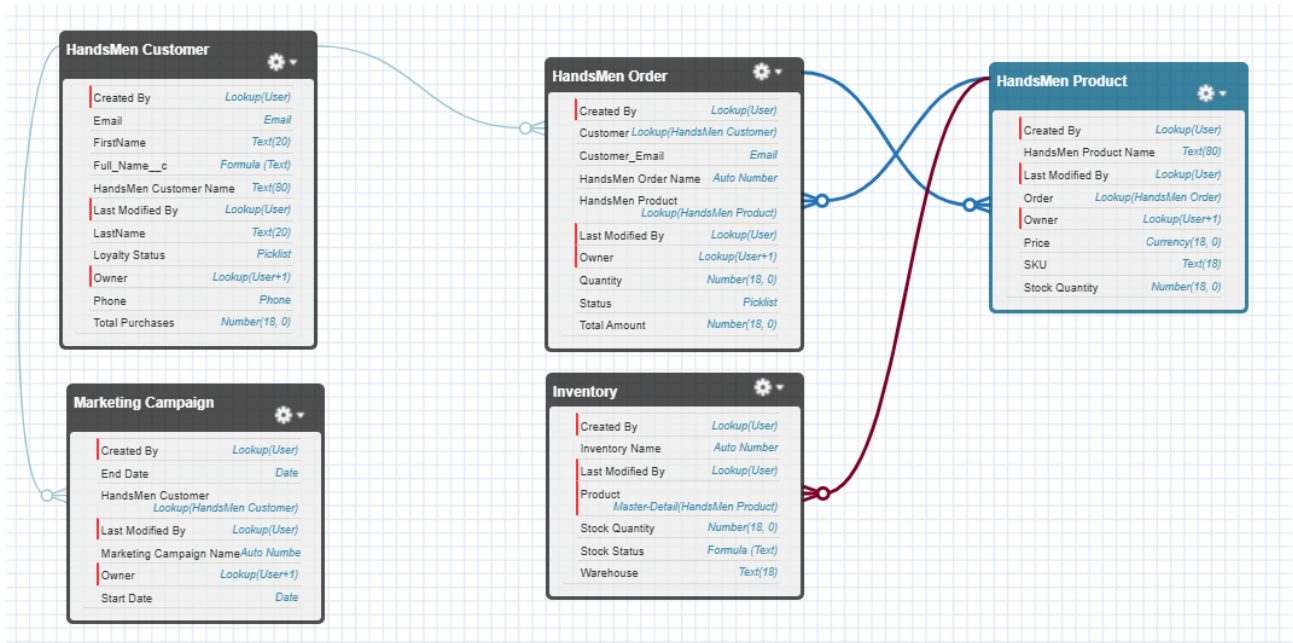


Figure 2 Schema Builder Data Model

4 Salesforce Configuration and Data Security

The project uses **custom tabs and a dedicated navigation workspace** in Lightning Experience, allowing users to easily access customer, product, order, inventory, and campaign records. Data access is managed through **profiles, roles, and permission sets**, ensuring users have appropriate permissions based on their responsibilities and business requirements.

Role	Intended Access
Sales Manager	Full access to customer and order records.
Inventory Manager	Read and edit access to inventory and product records.
Marketing Manager or Team	Read customer records and manage marketing campaign records.

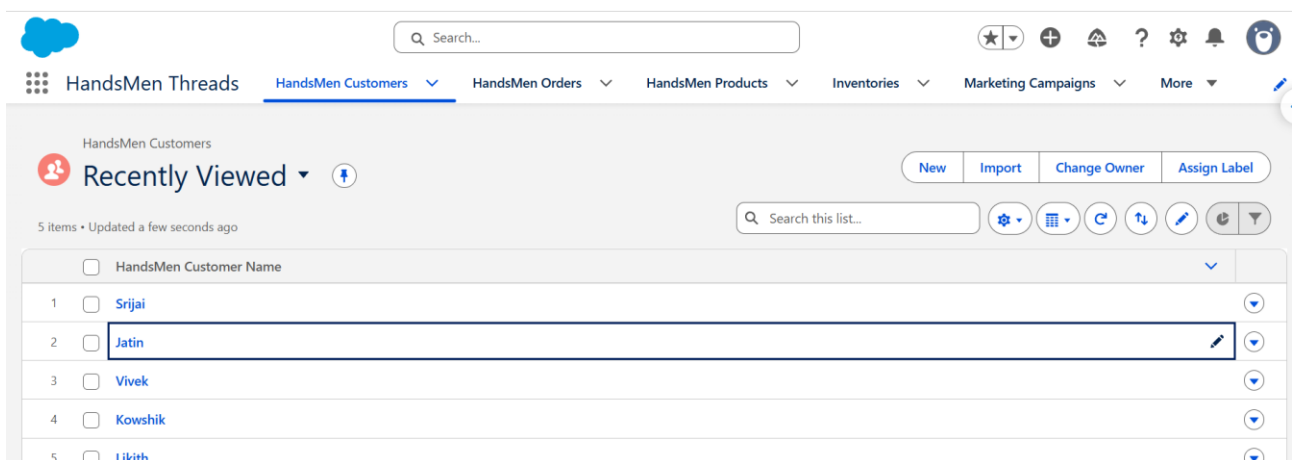


Figure 4 HandsMen Lightning App and Navigation



Figure 5 Role Hierarchy

4.1 Profiles Users and Permission Sets

Security Item	Configuration	Purpose
Profile	Platform 1 cloned from Standard User	Provides the base Salesforce Platform access.
User	Sales user assigned to Sales role	Supports customer and order activities.
User	Inventory user assigned to Inventory role	Supports product and inventory activities.
Permission Set	Permission Platform 1	Adds create, read, edit, and delete access for selected project objects.

Permission Sets

Permission Set: **Permission Platform 1**

Find Settings... Clone Edit Properties Manage Assignments View Summary

Permission Set Overview > Object Settings > HandsMen Customers

HandsMen Customers Edit

Tab Settings

Available	Visible
☐	☐ i

Object Permissions

Permission Name	Enabled
Read	☒
Create	☒
Edit	☒
Delete	☒
View All Records	☐
Modify All Records	☐
View All Fields	☐

Figure 1 HandsMen Customer permissions configured in Permission Platform 1

SETUP
Permission Sets

Permission Platform 1

[Permission Set Overview](#) >
 [Object Settings](#)
HandsMen Orders

HandsMen Orders

Tab Settings

Available	Visible
☐	☐

Object Permissions

Permission Name	Enabled
Read	☒
Create	☒
Edit	☒
Delete	☒
View All Records	☐
Modify All Records	☐
View All Fields	☐

Figure 2 HandsMen Order permissions configured in Permission Platform 1

SETUP
Permission Sets

Permission Platform 1

[Permission Set Overview](#) >
 [Object Settings](#)
HandsMen Products

HandsMen Products

Tab Settings

Available	Visible
☐	☐

Object Permissions

Permission Name	Enabled
Read	☒
Create	☒
Edit	☒
Delete	☒
View All Records	☐
Modify All Records	☐
View All Fields	☐

Figure 3 HandsMen Products permissions configured in Permission Platform 1

 SETUP Users						
☐	Action	Full Name ↑	Alias	Username	Role	Active Profile
☐	Edit	Chatter Expert	Chatter	chatty.00dfj00000qvy6veab.uqb75e177o6@chatter.salesforce.com		☒ Chatter Free User
☐	Edit	EPIC_OrgFarm	OEPIE	epic.1bdb2f992236@orgfarm.salesforce.com		☒ System Administrator
☐	Edit	Mikaelson, Daniel	dmika	238w1a05i5@vrsec.ac.in	Marketing	☒ Platform 1
☐	Edit	Mikaelson, Kol	kmika	238w1a05g5@vrsec.ac.in	Inventory	☐ Platform 1
☐	Edit	Mikaelson, Niklaus	nmika	238w1a05e5@vrsec.ac.in	Sales	☒ Platform 1
☐	Edit	Minnekanti, Likith Kowshik	lik	likithkowshik.cceffacabba7@agentforce.com		☒ System Administrator
☐	Edit	Minnekanti, Likith kowshik	lminn	likithkowshik@gmail.com		☐ VR User
☐	Edit	User_Integration	integ	integration@00dfj00000qvy6veab.com		☒ Analytics Cloud Integration User
☐	Edit	User_Security	sec	insightssecurity@00dfj00000qvy6veab.com		☒ Analytics Cloud Security User

Figure 4 Project Users and Assigned Roles

5 Data Quality and Business Rules

5.1 Formula Fields

Object	Formula Field	Return Type	Formula
HandsMen Customer	Full Name	Text	FirstName__c + " " + LastName__c
Inventory	Stock Status	Text	IF(Stock_Quantity__c > 10, "Available", "Low Stock")

5.2 Validation Rules

Object	Rule	Condition	Expected Message
HandsMen Order	Total Amount	Total_Amount__c <= 0	Please enter a correct amount.
Inventory	Stock Quantity	Stock_Quantity__c <= 0	Inventory count cannot be less than or equal to zero.
HandsMen Customer	Email	NOT(CONTAINS(Email__c, "@gmail.com"))	Please enter a correct Gmail address.

Setup

Home

Object Manager

SETUP > OBJECT MANAGER

HandsMen Order

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

ACTIVE ☒

Description

Error Condition Formula

Example: Discount_Percent__c > 0.30

More Examples...

Display an error if Discount is more than 30%

If this formula expression is **true**, display the text defined in the Error Message area

Insert Field

Use Formula Assistant

Insert Operator

Total_Amount__c <= 0

Check Syntax

No errors found

Functions

-- All Function Categories --

ABS

ACOS

ADDMONTHS

AND

ASCII

ASIN

Insert Selected Function

ABS(number)

Returns the absolute value of a number, a number without its sign

Help on this function

Figure 1 Total Amount field configuration and successful syntax validation

Setup Home Object Manager

SETUP > OBJECT MANAGER

HandsMen Customer

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Search Layouts

Rule Name:

Active: ☒

Description:

Error Condition Formula

Example: `Discount_Percent__c > 0.30` More Examples...

Display an error if Discount is more than 30%

If this formula expression is **true**, display the text defined in the Error Message area

Insert Field [Use Formula Assistant](#) Insert Operator

`NOT CONTAINS( Email__c , "@gmail.com")`

Check Syntax No errors found

Functions

-- All Function Categories --

ABS

ACOS

ADDMONTHS

AND

ASCII

ASIN

Insert Selected Function

ABS(number)

Returns the absolute value of a number, a number without its sign

[Help on this function](#)

Figure 2 Email formula field configuration and successful syntax validation

Setup Home Object Manager

SETUP > OBJECT MANAGER

Inventory

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Search Layouts

Rule Name:

Active: ☒

Description:

Error Condition Formula

Example: `Discount_Percent__c > 0.30` More Examples...

Display an error if Discount is more than 30%

If this formula expression is **true**, display the text defined in the Error Message area

Insert Field [Use Formula Assistant](#) Insert Operator

`Stock_Quantity__c <= 0`

Check Syntax No errors found

Functions

-- All Function Categories --

ABS

ACOS

ADDMONTHS

AND

ASCII

ASIN

Insert Selected Function

ABS(number)

Returns the absolute value of a number, a number without its sign

[Help on this function](#)

Figure 3 Stock Quantity configuration and successful syntax validation

New HandsMen Customer

* = Required Information

Information

* HandsMen Customer Name

Email

Enter a valid email address, such as name@email.com.

Phone

Loyalty Status

FirstName

LastName

Owner

Total Purchases

We hit a snag.

Review the following fields

- Email

Cancel Save & New Save

Figure 3 Customer email validation rule preventing an invalid email from being saved

6 Email Templates and Process Automation

6.1 Email Templates and Alerts

Template	Type	Used For
Order Confirmation	HTML	Sent when an order changes to Confirmed.
Low Stock Alert	Text	Sent to the inventory team when stock is below five.
Loyalty Program Email	HTML	Supports customer loyalty communication.

SETUP Classic Email Templates					
Edit Del	Appointment for Unauthenticated User using Engagement Channels-For third party.	Custom	✓	Email template for confirmation of an appointment when appointments are scheduled using engagement channels with third party video applications.	sfdcadm 6/1/2026
Edit Del	Canceled Service Appointment Confirmation Email	Custom	✓	Email Template to confirm canceling of a service appointment.	sfdcadm 6/1/2026
Edit Del	Commerce Reorder Portal Invitation	Custom	✓	Invite a contact to a Commerce Reorder Portal.	autoproc 6/1/2026
Edit Del	Group Service Appointments Enrollment Confirmation Email	Custom	✓	Email Template to confirm enrollment of an attendee to a Group service appointment.	sfdcadm 6/1/2026
Edit Del	Low Stock Alert	Text	✓		lik 9/11/2026
Edit Del	Loyalty Program Email	HTML	✓		lik 9/11/2026
Edit Del	Marketing Product Inquiry Response	Text	✓	Standard email response to website product inquiries	OEPIC 6/1/2026
Edit Del	Order Confirmation Email	HTML	✓		lik 9/11/2026

Figure 10 Email Templates List

SETUP Email Alerts				
All Email Alerts				
Email alerts are used to send emails from a flow or other automation.				
View: All Email Alerts Create New View				
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z Other				
New Email Alert				
Action	Description +	Email Template Name	Object	Last Modified Date
Edit Del	Low Stock Alert	Low Stock Alert	Inventory	9/11/2026
Edit Del	Loyalty program email	Loyalty Program Email	HandsMen Customer	9/11/2026
Edit Del	Order Confirmation Email Alert	Order Confirmation Email	HandsMen Order	9/11/2026

Figure 11 Order Confirmation Email Alert

6.2 Salesforce Flows

Flow	Type	Start Condition	Action
Order Confirmation Flow	Record Triggered	Order status changes to Confirmed	Sends the order confirmation email alert.
Stock Alert Flow	Record Triggered	Inventory stock quantity is below 5	Sends a low stock email alert.
Loyalty Status Update Flow	Schedule Triggered	Runs daily	Sets Gold, Silver, or Bronze from Total Purchases.

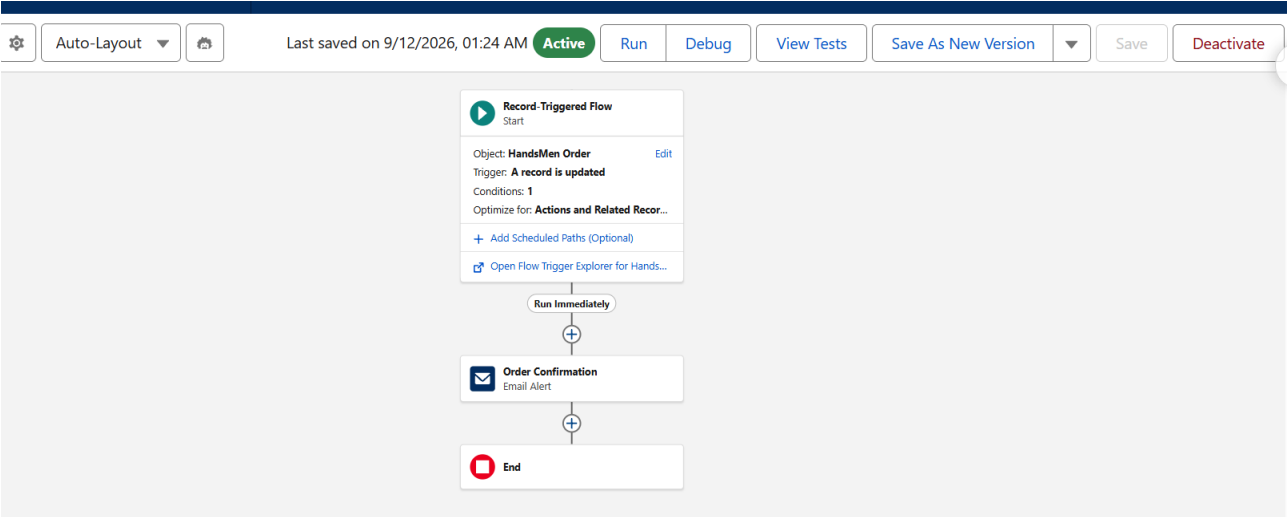


Figure 12 Order Confirmation Flow

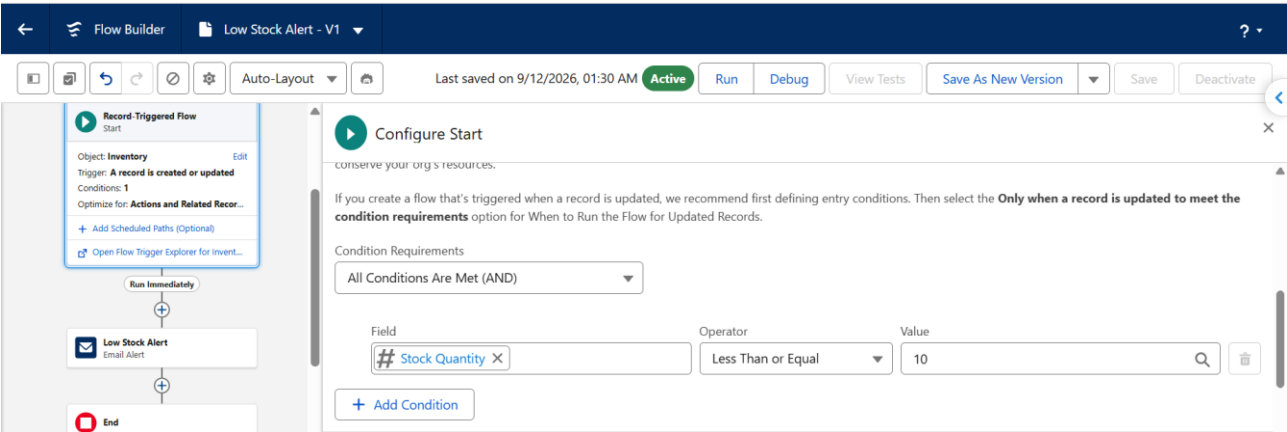


Figure 13 Stock Alert Flow

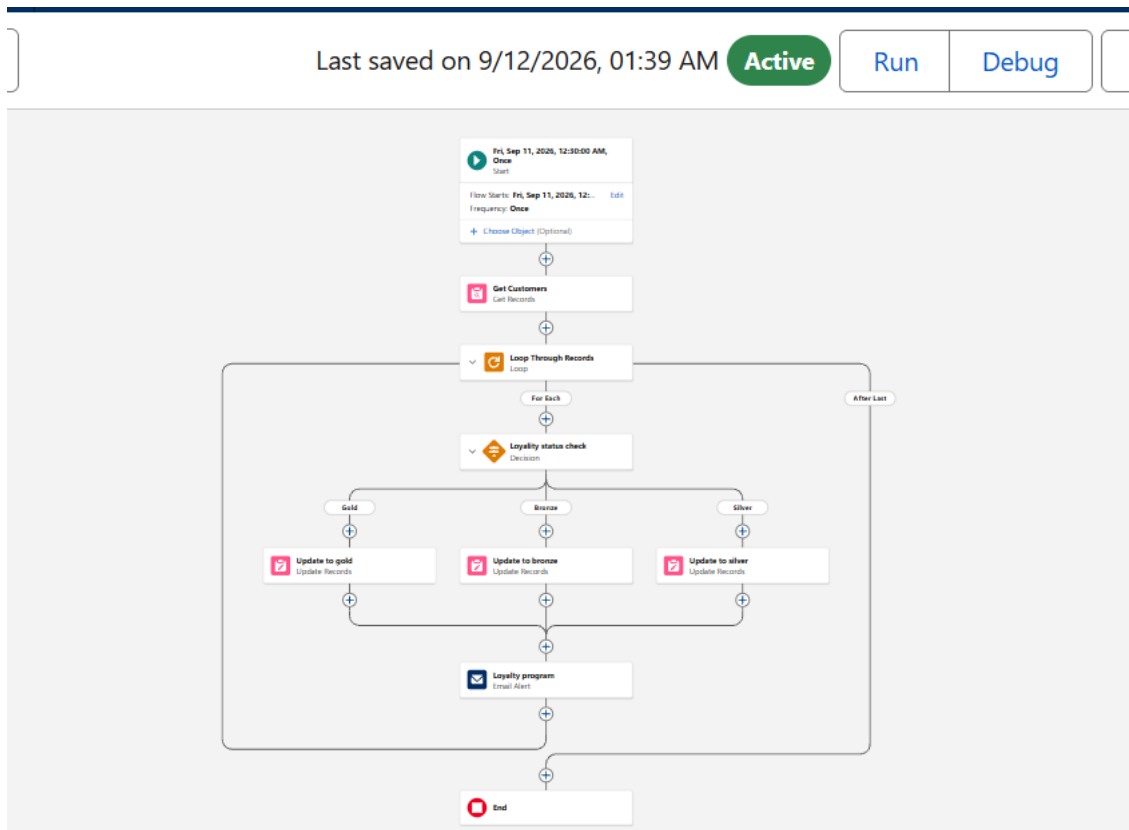


Figure 14 Loyalty Status Update Flow

7 Apex Triggers and Handler Logic

Apex is used for **transaction-level processing and reusable business logic** that cannot be handled effectively through standard Salesforce automation. The Order Trigger delegates status-based quantity validation to the **OrderTriggerHandler**, keeping the trigger lightweight while centralizing the business rule in a reusable and maintainable class.

Status	Quantity Rule	Outcome
Confirmed	Quantity must be greater than 500	The record is blocked when the quantity is missing or too low.
Pending	Quantity must be greater than 200	The record is blocked when the quantity is missing or too low.
Rejection	Quantity must equal 0	Rejected orders cannot contain a positive quantity.

```

trigger OrderTrigger on HandsMen_Order__c (before insert, before update) {
    OrderTriggerHandler.validateOrderQuantity(trigger.new);
}
  
```

```

public class OrderTriggerHandler {

    public static void validateOrderQuantity(List<HandsMen_Order__c> orderList) {

        for (HandsMen_Order__c order : orderList) {

            if (order.Status__c == 'Confirmed') {

                if (order.Quantity__c == null || order.Quantity__c <= 500) {

                    order.Quantity__c.addError('For Status "Confirmed", Quantity must be more than 500.');

```

Figure 15 OrderTriggerHandler Apex Class

```

trigger OrderTrigger on HandsMen_Order__c (before insert, before update) {

    if (Trigger.isBefore && (Trigger.isInsert || Trigger.isUpdate)) {

        OrderTriggerHandler.validateOrderQuantity(Trigger.new);

    }

}

```

Figure 16 OrderTrigger Apex Trigger

7.1 Order Total and Stock Deduction Triggers

Apex Artifact	Event	Responsibility
Order Total Trigger	Before insert and before update	Reads the related product price and sets Total Amount to Quantity multiplied by Price.
Stock Deduction Trigger	After insert and after update	When an order becomes Confirmed, totals ordered quantity by product and reduces Inventory Stock Quantity.

The stock deduction design collects product identifiers in a set, aggregates quantities in a map, queries matching inventory records once, and performs one bulk update. This avoids database operations inside the record loop.

```

trigger OrderTotalTrigger on HandsMen_Order__c (before insert, before update) {
    Set<Id> productIds = new Set<Id>();

    for (HandsMen_Order__c order : Trigger.new) {
        if (order.HandsMen_Product__c != null) {
            productIds.add(order.HandsMen_Product__c);
        }
    }

    Map<Id, HandsMen_Product__c> productMap = new Map<Id, HandsMen_Product__c>(
        [SELECT Id, Price__c FROM HandsMen_Product__c WHERE Id IN :productIds]
    );

    for (HandsMen_Order__c order : Trigger.new) {
        if (order.HandsMen_Product__c != null && productMap.containsKey(order.HandsMen_Product__c)) {
            HandsMen_Product__c product = productMap.get(order.HandsMen_Product__c);
            if (order.Quantity__c != null) {
                order.Total_Amount__c = order.Quantity__c * product.Price__c;
            }
        }
    }
}

```

Figure 17 Order Total Trigger

```

trigger StockDeductionTrigger on HandsMen_Order__c (after insert, after update) {
    Set<Id> productIds = new Set<Id>();
    for (HandsMen_Order__c order : Trigger.new) {
        Boolean shouldDeduct = false;

        if (order.Status__c == 'Confirmed') {
            if (Trigger.isInsert) {
                shouldDeduct = true;
            }
            else if (Trigger.isUpdate) {
                HandsMen_Order__c oldOrder = Trigger.oldMap.get(order.Id);

                if (oldOrder.Status__c != 'Confirmed') {
                    shouldDeduct = true;
                }
            }
        }

        if (shouldDeduct && order.HandsMen_Product__c != null) {
            productIds.add(order.HandsMen_Product__c);
        }
    }

    if (productIds.isEmpty()) {
        return;
    }

    // Deduct stock
    for (HandsMen_Order__c order : Trigger.new) {
        Boolean shouldDeduct = false;

        if (order.Status__c == 'Confirmed') {
            if (Trigger.isInsert) {
                shouldDeduct = true;
            }
            else if (Trigger.isUpdate) {
                HandsMen_Order__c oldOrder = Trigger.oldMap.get(order.Id);

                if (oldOrder.Status__c != 'Confirmed') {
                    shouldDeduct = true;
                }
            }
        }

        if (shouldDeduct &&
            order.HandsMen_Product__c != null &&
            order.Quantity__c != null) {

            Inventory__c existingInventory =
                inventoryMap.get(order.HandsMen_Product__c);

            if (existingInventory != null) {

                Decimal currentStock =
                    existingInventory.Stock_Quantity__c == null
                    ? 0
                    : existingInventory.Stock_Quantity__c;

                Decimal newStock =
                    currentStock - order.Quantity__c;

                inventoriesToUpdate.add(
                    new Inventory__c(
                        Id = existingInventory.Id,
                        Stock_Quantity__c = newStock
                    )
                );
            }
        }
    }

    if (!inventoriesToUpdate.isEmpty()) {
        update inventoriesToUpdate;
    }
}

```

Figure 18 Stock Deduction Trigger

8 Batch Apex and Scheduled Processing

The **InventoryBatchJob** implements both the `Database.Batchable` and `Schedulable` interfaces. It identifies products with low stock levels and processes inventory updates in manageable batches. The job can also be scheduled to run automatically, enabling regular and efficient inventory management without manual intervention.

Method or Process	Responsibility	Expected Evidence
start	Queries products that meet the low stock condition.	SOQL query and selected fields.
execute	Adds the restock quantity and updates records in bulk.	Loop, update list, and DML statement.
finish	Records that inventory synchronization completed.	Completion debug statement.
Schedulable execute	Starts the batch with a scope of 200.	Database <code>executeBatch</code> call.
Scheduled job	Runs Daily Inventory Sync automatically.	Scheduled Jobs entry and next run time.

```

global class InventoryBatchJob implements Database.Batchable<SObject>, Schedulable {
    global Database.QueryLocator start(Database.BatchableContext BC) {
        return Database.getQueryLocator(
            'SELECT Id, Stock_Quantity__c FROM Product__c WHERE Stock_Quantity__c < 10'
        );
    }

    global void execute(Database.BatchableContext BC, List<SObject> records) {
        List<HandsMen_Product__c> productsToUpdate = new List<HandsMen_Product__c>();

        // Cast SObject list to Product__c list
        for (SObject record : records) {
            HandsMen_Product__c product = (HandsMen_Product__c) record;

            product.Stock_Quantity__c += 50; // Restock logic
            productsToUpdate.add(product);
        }

        if (!productsToUpdate.isEmpty()) {
            try {
                update productsToUpdate;
            } catch (DmlException e) {
                System.debug('Error updating inventory: ' + e.getMessage());
            }
        }
    }

    global void finish(Database.BatchableContext BC) {
        System.debug('Inventory Sync Completed');
    }

    // Scheduler Method
    global void execute(SchedulableContext SC) {
        InventoryBatchJob batchJob = new InventoryBatchJob();
        Database.executeBatch(batchJob, 200);
    }
}

```

Figure 19 InventoryBatchJob Apex Class


SETUP
Scheduled Jobs

Percentage of Scheduled Jobs Used: 2%

You have currently used 2 scheduled Apex jobs out of an allowed organization limit of 100 active or scheduled jobs. To learn about how this limit is calculated and what contributes to it see the [Lightning Platform Apex Limits](#) topic.

View: [All Scheduled Jobs](#) [Create New View](#)

A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | Other | **All**

[Schedule Apex](#)

Action	Job Name ↑	Submitted By	Submitted	Started	Next Scheduled Run	Type	Cron Trigger ID
Manage Del Pause Job	Daily Inventory Sync	Minnekanti, Likith Kowshik	9/11/2026, 1:41 PM	9/17/2026, 12:02 AM	9/18/2026, 12:00 AM	Scheduled Apex	08efj00000aD8j1

Figure 20 Scheduled Inventory Job

9 User Interface and Sample Records

The user interface is designed to provide business users with easy access to configured fields, relationships, and automated processes through **Salesforce Lightning Experience**. Sample records use realistic, non-sensitive data to demonstrate the functionality of the system. The browser zoom is maintained between **80% and 100%** to ensure that labels, field values, and page components remain clearly visible during demonstrations.

Record	Recommended Fields to Show
HandsMen Customer	Full name, email, phone, loyalty status, total purchases
HandsMen Product	Product name, SKU, price, stock quantity
HandsMen Order	Order number, customer, product, status, quantity, total amount
Inventory	Inventory number, product, warehouse, stock quantity, stock status
Marketing Campaign	Campaign number or name, dates, related customer

The screenshot displays two side-by-side record views in the HandsMen Threads Salesforce Lightning Experience. The left view shows a 'HandsMen Customer' record for 'Likith', and the right view shows a 'HandsMen Product' record for 'T-Shirt Cloth'. Both records are owned by 'Likith Kowshik Minnekanti'.

HandsMen Customer Record (Likith):

- HandsMen Customer Name:** Likith
- Email:** likithkowshik@gmail.com
- Phone:** 08985572848
- Loyalty Status:** Bronze
- FirstName:** Likith
- LastName:** Minnekanti
- FullName:** Likith Minnekanti

HandsMen Product Record (T-Shirt Cloth):

- HandsMen Product Name:** T-Shirt Cloth
- Order:** O-0002
- SKU:** TSHIRT001
- Price:** \$3
- Stock Quantity:** 100
- Created By:** Likith Kowshik Minnekanti, 9/11/2026, 10:09 AM
- Last Modified By:** Likith Kowshik Minnekanti, 9/17/2026, 11:50 PM

Figure 21 Customer and Product Sample Records



HandsMen OrderNumber

O-0002

Owner

 [Likith Kowshik Minnekanti](#)

Customer

[Likith](#)

Status

Confirmed



Quantity

509



Total Amount

1,527



HandsMen Product

[T-Shirt Cloth](#)

Customer email

likithkowshik0215@gmail.com

Figure 22 Order and Inventory Sample Records

10 Functional Testing Evidence

Functional testing should show both configuration and business outcomes. Capture the action and its result, and use before and after screenshots where a value changes.

Test Case	Action	Expected Result	Evidence
Order quantity validation	Save a status and quantity combination that violates the handler rule.	Save is blocked with the Apex error.	Error on Order page
Order total	Create or edit an order with product and quantity.	Total Amount equals Quantity multiplied by Price.	Order values
Stock deduction	Change an order from Pending to Confirmed.	Inventory decreases once by the ordered quantity.	Before and after stock
Order confirmation	Confirm an order.	Customer receives the confirmation email.	Email inbox
Low stock alert	Set stock below 5.	Inventory manager receives an alert.	Email inbox
Loyalty update	Run or debug the scheduled flow with purchase values.	Customer loyalty changes to Gold, Silver, or Bronze.	Flow debug and record
Batch processing	Execute InventoryBatchJob.	Low stock records are processed and the job completes.	Apex Jobs and record

HandsMen Order

O-0002

HandsMen OrderNumber
O-0002

Owner
 Likith Kowshik Minnekanti

Customer

Likith

Status

Confirmed

Quantity

500

For Status "Confirmed", Quantity must be greater than 100.

Total Amount

1,527

We hit a snag.

Review the following fields

- Quantity

10.1 Automation Test Result

Figure 24 Order Total Calculation

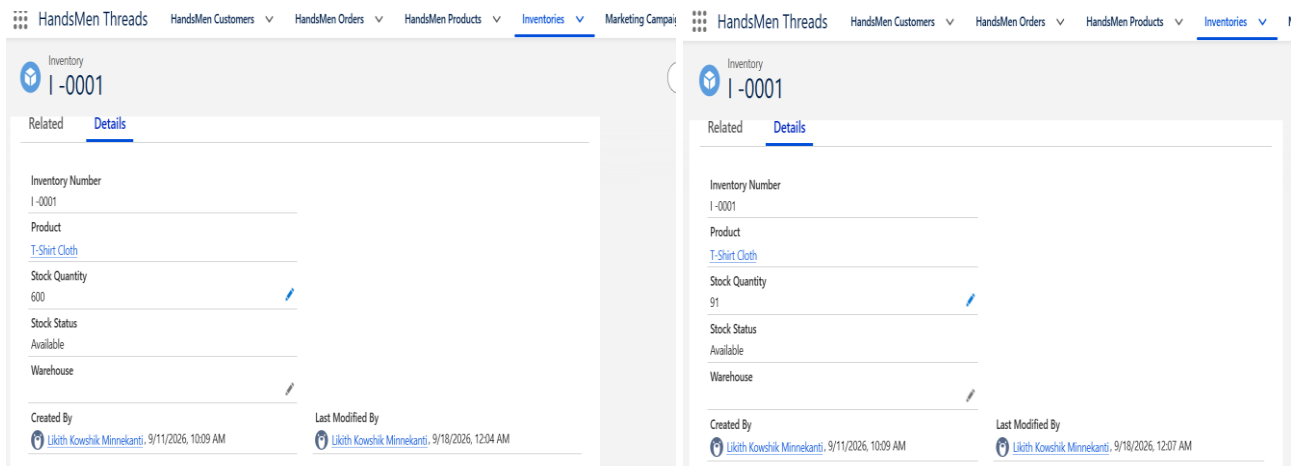
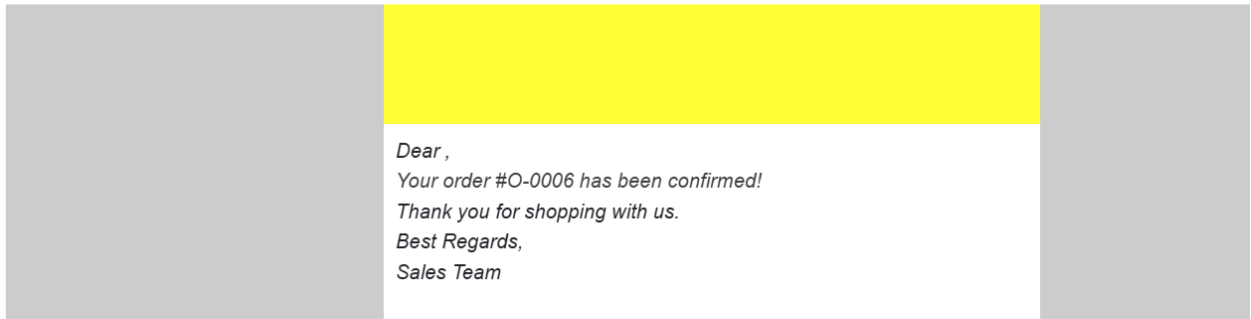


Figure 25 Inventory Before and After Confirmation

Your Order has been Confirmed! [Inbox x](#)

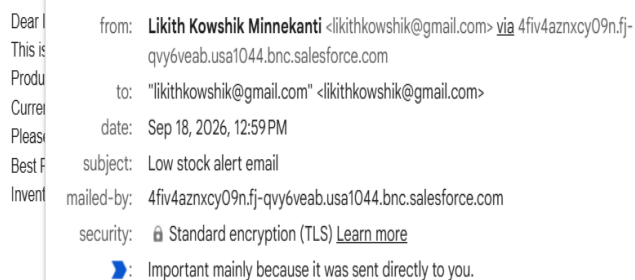
Likith Kowshik Minnekanti via [9kd90mv10o1l0w.fj-qvy6veab.usa1044.bnc.salesforce.com](#)
to me

1:10 PM (0 minutes ago)



Low stock alert email [Inbox x](#)

Likith Kowshik Minnekanti via [4fiv4aznxcy09n.fj-qvy6veab.usa1044.bnc.salesforce.com](#)
to me



Low stock alert email [Inbox x](#)

Likith Kowshik Minnekanti via [4fiv4aznxcy09n.fj-qvy6veab.usa1044.bnc.salesforce.com](#)
to me

Dear Inventory Manager,
This is to inform you that the stock for the following product is running low:
Product Name: T-Shirt Cloth
Current Stock Quantity: 7
Please take the necessary steps to restock this item immediately.
Best Regards,
Inventory Monitoring System

Figure 26 Automated Email Evidence

10.2 Scheduled Automation and Batch Results

Likith	
Email	likithkowsnik@gmail.com
Phone	08985572848
Loyalty Status	Bronze
FirstName	Likith
LastName	Minnekanti
FullName	Likith Minnekanti
Total Purchases	500

Figure 27 Loyalty Status Update Result

Apex Jobs

[Click here to go to the new batch jobs page](#)

Apex Jobs

Monitor the status of all Apex jobs, and optionally, abort jobs that are in progress.

Percent of Asynchronous Apex Used: 0%
 You have currently used 3 asynchronous Apex operations out of an allowed 24-hour organization limit of 250,000. To learn about how this limit is calculated and what contributes to Lightning Platform Apex Limits topic.

View: All [Create New View](#)

Action	Submitted Date	Job Type	Status	Status Detail	Total Batches	Batches Processed	Failures	Submitted By	Completion Date	Apex Class	Apex Method
	9/18/2026, 1:02 AM	Batch Apex	Completed		1	1	0	Minnekanti, Likith Kowshik	9/18/2026, 1:03 AM	InventoryBatchJob	

Figure 28 Batch Job Execution Result

11 Conclusion

The **HandsMen Threads Salesforce project** provides an integrated CRM and operations solution tailored for a fashion business. It centralizes customer, product, order, inventory, and campaign data while using validation rules, formula fields, security controls, and automation to improve business processes.

The project combines **Salesforce configuration, Flow automation, Apex development, and data security** to implement practical business requirements such as order processing, customer communication, loyalty management, and inventory control. The use of Apex triggers, Batch Apex, and Scheduled Apex further demonstrates the application of scalable business logic and bulk processing.

Overall, the project demonstrates key **Salesforce Platform Developer skills** through a complete implementation covering data modeling, security, automation, development, and testing. The supporting screenshots and functional evidence provide a clear view of the solution from its initial configuration to its final business outcomes.